

Experiment-8

Q1. Write a c program volume of a capsule-shaped solid made of a solid cylinder and two hemispheres when radius and height are given.

(A) Understand the Problem:

Problem Statement:

Write a C program to find the volume of a capsule-shaped solid made up of a cylinder and two hemispheres when the radius and height of the cylindrical part are given.

Input:

- Radius of capsule, "r"
- Height of cylindrical part, "h"

Output:

- Volume of the capsule

Formula:

A capsule consists of:

- One cylinder
- Two hemispheres, which together form one sphere

Volume of cylinder:

$$V_c = \pi r^2 h$$

Volume of two hemispheres:

$$V_s = \frac{4}{3} \pi r^3$$

Therefore,

$$\text{Volume of capsule} = \pi r^2 h + \frac{4}{3} \pi r^3$$

For the program, take $\pi = 3.14159$.

(B)Test Case Table:

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	r = 2, h = 5	Volume = 87.96	Volume = 87.96	Pass
2	r = 3, h = 10	Volume = 395.84	Volume = 395.84	Pass
3	r = 1, h = 6	Volume = 23.04	Volume = 23.04	Pass

(C)Algorithm:

1. Start.
2. Declare variables "radius", "height", and "volume" as float.
3. Ask the user to enter the radius.
4. Read the value of "radius".
5. Ask the user to enter the height of the cylindrical part.
6. Read the value of "height".
7. Calculate the volume using:
$$\text{volume} = 3.14159 \times \text{radius} \times \text{radius} \times \text{height} + (4.0/3.0) \times 3.14159 \times \text{radius} \times \text{radius} \times \text{radius}$$
8. Display the calculated volume.
9. Stop.

Algorithm Evaluation

For Test Case 1, radius = 2 and height = 5. The algorithm calculates the volume as 87.96.

For Test Case 2, radius = 3 and height = 10. The algorithm calculates the volume as 395.84.

For Test Case 3, radius = 1 and height = 6. The algorithm calculates the volume as 23.04.

Therefore, the algorithm gives the expected result for all three test cases.

(D)Program, compilation and execution:

```
kali@kali: ~  
$ nano solid.c  
$ gcc solid.c -o solid  
/usr/bin/x86_64-linux-gnu-ld.bfd: cannot find solid.c-o: No such file or directory  
/usr/bin/x86_64-linux-gnu-ld.bfd: cannot find solid: No such file or directory  
collect2: error: ld returned 1 exit status  
$ ./solid  
./solid: command not found  
$ gcc solid.c -o solid  
$ ./solid  
Enter the radius of the capsule: 2  
Enter the height of the cylindrical part: 5  
Volume of the capsule = 96.34  
$
```

```
GNU nano 9.0 solid.c  
#include <stdio.h>  
  
int main()  
{  
    float radius, height, volume;  
    float pi = 3.14159;  
  
    printf("Enter the radius of the capsule: ");  
    scanf("%f", &radius);  
  
    printf("Enter the height of the cylindrical part: ");  
    scanf("%f", &height);  
  
    volume = pi * radius * radius * height  
            + (4.0 / 3.0) * pi * radius * radius * radius;  
  
    printf("Volume of the capsule = %.2f\n", volume);  
    return 0;  
}
```

Example Execution:

Enter the radius of the capsule: 2
Enter the height of the cylindrical part: 5
Volume of the capsule = 87.96

The program can then be executed using the remaining test cases.

Test Case	Expected Output	Actual Output	Result
1	87.96	87.96	Pass

2| 395.84| 395.84| Pass

3| 23.04| 23.04| Pass

Thus, the program successfully passes all the prepared test cases.

Question 2: Write a c program to find third angle of a triangle when other two angles are given.

(A) Understand the Problem:

Problem Statement:

Write a C program to find the third angle of a triangle when two angles are given.

Input:

- First angle, "angle1"
- Second angle, "angle2"

Output:

- Third angle of the triangle

The sum of the three interior angles of a triangle is 180° .

Therefore:

Third angle = $180^\circ - (\text{First angle} + \text{Second angle})$

(B) Test Case Table:

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	angle1 = 60° , angle2 = 60°	Third angle = 60°	Third angle = 60°	Pass
2	angle1 = 90° , angle2 = 45°	Third angle = 45°	Third angle = 45°	Pass
3	angle1 = 70° , angle2 = 50°	Third angle = 60°	Third angle = 60°	Pass

Step 3: Algorithm

1. Start.
2. Declare integer variables "angle1", "angle2", and "angle3".
3. Ask the user to enter the first angle.
4. Read "angle1".
5. Ask the user to enter the second angle.
6. Read "angle2".
7. Calculate the third angle using:
$$\text{angle3} = 180 - (\text{angle1} + \text{angle2})$$
8. Display "angle3".
9. Stop.

Algorithm Evaluation

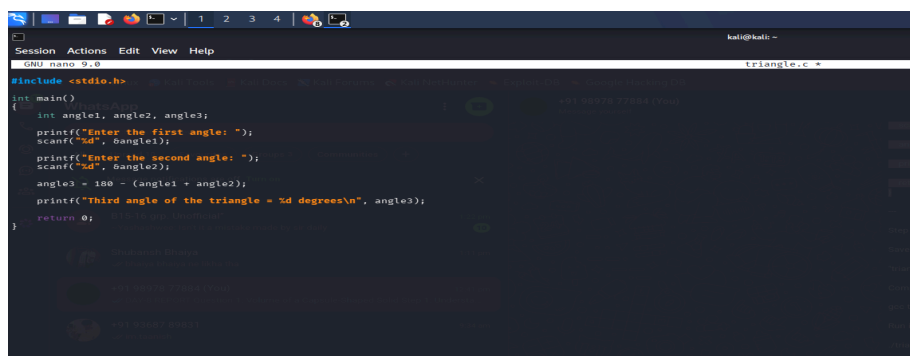
For Test Case 1, angle1 = 60° and angle2 = 60°. The algorithm calculates angle3 = 60°.

For Test Case 2, angle1 = 90° and angle2 = 45°. The algorithm calculates angle3 = 45°.

For Test Case 3, angle1 = 70° and angle2 = 50°. The algorithm calculates angle3 = 60°.

Therefore, the algorithm produces the expected result for all three test cases.

(D)Program, compile and execution:



```
Session Actions Edit View Help
GNU nano 9.0
triangle.c
#include <stdio.h>
int main()
{
    int angle1, angle2, angle3;
    printf("Enter the first angle: ");
    scanf("%d", &angle1);
    printf("Enter the second angle: ");
    scanf("%d", &angle2);
    angle3 = 180 - (angle1 + angle2);
    printf("Third angle of the triangle = %d degrees\n", angle3);
    return 0;
}
```

```
(kali@kali)-[~]
$ nano triangle.c
(gcc) 98978 77884 (You)
(gcc) 93687 89831
$ gcc triangle.c -o triangle
(gcc) 93687 89831
$ ./triangle
Enter the first angle: 37
Enter the second angle: 53
Third angle of the triangle = 90 degrees
(gcc) 93687 89831
$
```

Example Execution:

Enter the first angle: 60

Enter the second angle: 60

Third angle of the triangle = 60 degrees

Evaluate the program using all the prepared test cases:

Test Case	Expected Output	Actual Output	Result
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1	60°	60°	Pass
---	-----	-----	------

2	45°	45°	Pass
---	-----	-----	------

3	60°	60°	Pass
---	-----	-----	------

Therefore, the C program passes all three test cases successfully.

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